ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

Project ID: 25-26J-106

Project Proposal Report

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HUMAN-LED SELECTION THROUGH GAZE-GUIDED ROBOTIC ASSISTANCE

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Declaration of the candidate & Supervisor

I declare that this is my own work, and this dissertation does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any other university or institute of higher learning and, to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due acknowledgement is made in the text.

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Abstract

In Human-Robot Collaboration (HRC), intuitive interaction methods play a critical role in allowing natural and efficient interaction between humans and robots to convey their intentions to each other. Gaze interaction is one such method that shows promise for its use in assistive applications that include differently-abled people. Nonetheless, gaze-controlled systems currently available in the market depend heavily on eye-tracking devices that are costly and unaffordable for ordinary users. In this research, a computer vision-based method for estimating the gaze direction for HRC applications is designed without the need for specialized hardware.

In particular, the proposed system combines vision-based gaze estimation, region-based intention detection, and robotic arm control into a single assistive system. The data set consists of images from RGB and CCTV cameras showing faces that represent seven pre-defined gaze regions. The proposed deep learning model is based on ResNet-18 architecture, which was trained on the created dataset for gaze direction estimation. The performance of the proposed solution was evaluated based on accuracy, precision, recall, F1-score, and confusion matrix metrics.

Gaze regions were mapped to pre-defined workspace locations of objects, providing gaze-based robotic object selection using a robotic arm. Gaze estimation with robotic manipulation illustrated that this concept was possible to realize and convert visual intentions into robotic actions. Results showed that gaze perception without hardware could serve as a cost-effective solution compared to traditional eye tracking systems, offering an intuitive interface for human–robot collaboration.

The study makes contributions to gaze guidance in human–robot collaboration through its use of intelligent perception along with robotic actions for assistive selection. The presented system offers future possibilities for applications in assistive robotics, user interface design for computers, and human–robot collaboration.

Keywords: Gaze Direction Estimation, Human–Robot Collaboration, Assistive Robotics, Deep Learning, Region-Based Gaze Classification.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
CCTV	Closed-Circuit Television
CSV	Comma-Separated Values
DL	Deep Learning
F1-Score	Harmonic Mean of Precision and Recall
GPU	Graphics Processing Unit
HRC	Human–Robot Collaboration
HRI	Human–Robot Interaction
IoT	Internet of Things
ML	Machine Learning
RGB	Red, Green, Blue
ReLU	Rectified Linear Unit
ROI	Region of Interest
RNN	Recurrent Neural Network
ResNet	Residual Neural Network
UI	User Interface
UR	Upper Right
UC	Upper Center
UL	Upper Left
LR	Lower Right

LC	Lower Center
LL	Lower Left
YOLO	You Only Look Once
FPS	Frames Per Second
API	Application Programming Interface
GUI	Graphical User Interface
CPU	Central Processing Unit
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

1. INTRODUCTION

1. 1 Background & Literature survey

Human-Robot Collaboration (HRC) is an innovative area of research that aims at ensuring the ability of humans and robots to work together in a safe and efficient manner. In contrast to industrial robots that can function in isolation under pre-programmed conditions without much contact with humans, collaborative robots, also known as cobots, are robots that exist in a joint environment with humans and assist humans by cooperating with them and adapting their performance accordingly. The emergence of collaboration as opposed to automation has paved many ways to the utilization of HRC technology, particularly in situations where intention recognition is critical for collaboration [1], [2].

One of the main goals of HRC involves creating methods of interaction between humans and robots where the former can understand the intention of the latter in an intuitive way. Traditional forms of human–machine interaction including mechanical controllers, keyboards, touch screens, and other devices requiring significant physical strength or abilities do not fit users with limited movement. Therefore, there has been much recent research activity devoted to studying natural human–robot interfaces which help improve accessibility and inclusiveness by avoiding additional cognitive or physical burdens [3].

Gaze tracking is one of the human–robot interaction modalities receiving particular attention among numerous other intuitive forms of communication. Human gaze is a strong indicator of attention and intention in the process of communication. In everyday communication, gaze serves as an important means of attracting attention, focusing on some aspect of communication, and coordinating actions between participants [4].

In particular, for people who have disabilities or are physically incapable of controlling any of the above devices, the use of gaze as a different method of communication and manipulation is possible, since these conventional interfaces may prove too difficult to operate. With gaze interaction, the user does not necessarily need to move his/her body or hands but instead communicate the intention through the eyes, which may provide a way for the robot to help in completing certain tasks by itself. Thus, gaze-based interaction has

become a rather interesting subject matter for studying assistive robotics and inclusive human-computer interaction [5], [6].

The first implementations of gaze-based interaction involved the use of specialized devices such as infrared eye tracking technology, wearable eye-tracking devices, electrooculographic sensors, and special-purpose eye tracking cameras. All of these approaches were highly accurate, thus, making them popular in fields such as usability studies and rehabilitation. Yet, while these methods provided a great level of accuracy, they had other disadvantages that limited their applicability. In particular, they could be relatively expensive and require precise lighting conditions, extensive calibration, and specific devices [7], [8].

This has led to increased research interests in gaze estimation techniques that use conventional RGB cameras and computer vision methods instead of eye tracking hardware. The recent developments in computer vision and deep learning have resulted in considerable advancements in the area. Appearance based gaze estimation, facial landmarks detection, head pose estimation, and deep learning networks have shown promise in predicting gaze direction using regular cameras available on laptops, smart devices, and embedded vision platforms [9], [10].

Among these techniques, Convolutional Neural Networks have greatly contributed towards the improvements in gaze estimation accuracy when implemented through software solutions. Convolutional neural network architectures allow the system to learn effective visual features from facial regions, eye regions, and head poses to estimate gaze direction. Deep learning architectures like ResNet are employed for extracting spatial features due to their effective learning techniques, whereas lighter architectures like MobileNet make inference easy and fast [11], [12].

In the same vein, deep feature extraction approaches based on Caffe have also been utilized in extracting eye-region features, and appearance-based approaches have been applied to the prediction of gaze using face features. Apart from the continuous approach to gaze estimation, there have been approaches to discrete gaze direction prediction where,

instead of estimating the exact position of gaze, it is classified to spatial regions such as left, right, center, up, or down regions. This is especially useful in robotic tasks of selecting objects where precise point-of-gaze estimation is less important than accurate intention detection [13], [14].

There has been an increased interest lately in the use of gaze information in collaborative robotics. Applications of gaze in collaborative robotics, especially human-robot collaboration (HRC) scenarios, have included using gaze for recognizing intentions of humans, situational awareness of humans within a common workspace, anticipation of robot behavior by humans, and handing over objects between humans and robots, among others. By using gaze information, robots can determine where the attention of their human collaborators lies and what action the human intends to take next [15], [16].

Nonetheless, while these advances have been made, most current research on the topic relies on the use of gaze as either contextual information or as a monitoring tool without using it to generate commands to execute tasks. Most of the research focuses on how gaze can be used as part of the process of intention recognition and even in helping to predict the intentions of the user; however, it does not go further into how this gaze can translate into specific instructions to be carried out by robots.

This particular problem is even more relevant when applied to situations in which assistive robotics technology is used to help people who are differently abled, where gaze might be a key input source rather than a supplemental one. Gaze-based control in assistive robotics has great potential in increasing the independence of differently abled individuals. However, to do so, a number of issues need to be addressed.

These issues have been addressed by suggesting the software-only solution for human-robot collaboration for assistive object selection using eye gaze. This study goes beyond other literature works that concentrated on either gaze detection/classification, as it uses gaze estimation to create a full-fledged perception-to-action workflow, where visual intention is converted to robot actions.

The suggested approach allows the differently abled student to simply look at a particular spatial area in the workspace for selecting the intended object. Gaze estimation is done through a vision-based system, making use of deep learning techniques and computer vision concepts. The gaze direction prediction made from the aforementioned vision-based system is mapped to the pre-defined spatial regions in the workspace, associated with different objects. Once this mapping is done, the object located within the mapped region is selected by the robot.

This results in a novel gaze-guided HRC approach that combines perception, intention interpretation, and robotic manipulation in one system. Instead of considering gaze estimation and robotic manipulation separately as two distinct problems, both components become integral parts of the same assistive interaction process.

The importance of a framework like this is evident from the contribution in different research areas. First, concerning assistive robotics, a gaze-guided interaction provides an accessible means of manipulating objects. The possibility of robotic assistance using only the gaze might provide new opportunities for independent execution of daily tasks and education or therapy in an assistive setting [17].

Second, in terms of inclusive human-computer interaction research, the contribution is made by providing an example of an interface based on universal design considerations. The design assumes that the use of standard control mechanisms is not guaranteed and considers interaction grounded on natural user behavior.

Third, in the context of vision-based HRC systems, the importance is related to the contribution to intention-aware collaborative robotic systems. As intelligent collaboration becomes increasingly widespread in HRI, the idea of integrating intention interpretation in collaborative robotics seems to be very promising [18].

Fourth, the contribution made by the proposed framework is in the area of low-cost technologies for improving accessibility through the use of standard vision sensors and pure software solutions that do not require costly specialized equipment.

Lastly, by integrating the gaze perception component with the robot's capabilities to manipulate objects, we can develop intelligent robotic selection systems, wherein the perception-based interaction leads to a better understanding of user actions.

From the perspective of computer science, another contribution to the literature can be identified through an emerging trend towards the integration of computer vision, deep learning, spatial reasoning, and robotics into a single assistive system framework.

The importance of this research is made even more relevant thanks to the latest advances in both collaborative robotics and artificial intelligence. Contemporary collaborative robots feature perceptual capacities that allow them to interact with humans in safe manners and perform tasks in human-shared spaces. Moreover, due to the latest advances in deep learning, it became possible to implement software-based gaze estimation even using inexpensive equipment. In other words, there is an opportunity to explore gaze-guided robotic assistants that were not practical before.

However, implementing gaze-guided robotic assistants is difficult. Estimating gazes using only software requires dealing with differences in head poses, faces, lightings, and behavior of the person whose gaze is being estimated. The estimated gaze needs to be sufficiently accurate to enable further object selection based on the results of such estimation. Also, region mapping will be required to convert visual intent into actions in the workspace. Finally, controlling robotic manipulators is a complex task.

Dealing with all these issues is what motivated the main goal of this dissertation – creating and testing the system for controlling a collaborative robotic arm to select objects using gaze direction only.

Through exploration of the issue at hand, the study will make a contribution both theoretical and practical towards the development of assistive HRC. On one hand, the theoretical aspect lies in the attempt of looking at the possibility of using gaze as a means through which human intention can be detected. On the other hand, the study will help in

making progress on development of assistive robotics technologies that would be accessible and affordable.

Finally, the proposed research framework lies within a wider context of vision of intelligent human-robot collaboration. This vision focuses on ability of a robot to understand and execute human intention in an effective manner. Through conversion of gaze into robotic action, the study will make contribution towards realizing this vision.

1. 2 Research Gap

However, several critical gaps in research persist concerning the design of gaze-driven assistive robots that are practical and feasible for use in reality. While the previous studies made considerable advances in the field of gaze detection, intention-driven robotics, and human-assisted robot interaction, there is still much room left for improvement regarding the availability, system integration, inclusion, and overall system implementation.

Firstly, many of the gaze-driven systems require expensive eye trackers and other costly devices, limiting their scalability and feasibility in assistive settings. Although commercial and experimental eye trackers have a number of advantages in terms of providing accurate and reliable gaze information, their necessity for the presence of specialized hardware, calibration procedures, and controlled operational conditions makes them less feasible to implement across all assistive contexts [19], [20]. Thus, a serious gap persists in the development of feasible and efficient gaze-driven systems capable of operating without specialized hardware.

Secondly, even though gaze estimation using computer vision and deep learning methods have come a long way, there seems to be an over-emphasis on the aspect of gaze classification and the improvements in gaze models themselves. Most researches have been carried out based upon the principle that gaze estimation is a separate problem for computer vision and there is more emphasis on how accurate the gaze classification is, or what angular error occurs in estimating the gaze [21], [10]. This brings forth the point that research is lacking in terms of gaze estimation in human–robot interaction, where gaze is used to control robotic tasks.

Yet another major research gap involves the lack of research conducted on gaze-controlled object selection in assistive human–robot interaction. Despite considerable research on gaze interaction in general-purpose HCI systems and robotics applications, little is known about enabling differently-abled individuals to operate assistive robot functions using gaze interaction alone [6], [22]. It appears that the majority of the current literature on the matter focuses on general collaborative tasks, user attention, or multimodal interactions rather than developing user-focused and accessibility-oriented assistive robotics frameworks.

A similar gap emerges from the prevalent use of multimodal interaction interfaces in most robotic interaction systems based on the use of speech recognition, hand gesture recognition, body-mounted devices, manual control, and other types of inputs. While multimodal interaction is helpful in certain circumstances, such a paradigm does not appear very helpful for differently-abled users who cannot speak or move their body parts effectively. For this reason, research on the development of hands-free and easy-to-use interaction interfaces in which gaze control suffices appears to be needed [23], [16].

A final notable gap is that related to system integration. While each of the elements in question has been explored individually in their own right, namely software-only gaze estimation, spatial region recognition, object localization, and robotic manipulation, very few works integrate the above-mentioned components into an integrated end-to-end pipeline for assistive object manipulation. Existing approaches tend to address the issues at hand separately from each other, thus failing to provide an integrated perception-to-action mechanism for translating visual intention into robotic action, thereby leaving a substantial gap to fill [24], [18].

Additionally, the problem of software-only gaze estimation in its current state leaves open several unresolved issues associated with model robustness and practical applicability. While many of these methods are developed on datasets and tested under controlled conditions, most of them fail in practical scenarios with variable light conditions, varying head poses, diverse users, and uncertain gaze locations. To make use of gaze as a practical robotic control mode, it is necessary to address some of these issues [14], [25].

One more unexplored topic is the role of gaze as a direct control command rather than a contextual signal. While collaborative robots use gaze cues for intention prediction, shared attention modeling, and robot anticipation, they do not exploit gaze directly as a command

for initiating robotic manipulation. The difference is critical, as exploiting gaze as a direct control means dealing with issues related to intention confirmation, command initiation, and action execution robustness. Little research has been done in terms of the use of gaze control command in assistive collaborative robotics [4].

Another issue concerns the development of low-cost accessible technologies combining artificial intelligence and collaborative robotics for differently-abled people. Despite the importance of such technology both socially and technologically, most advanced assistive robotic systems are too expensive and specialized for widespread application. Therefore, affordable, vision-enabled, assistive robotic systems are still underdeveloped [26].

Together, these shortcomings suggest the need for research that goes beyond the incremental improvements in gaze tracking technology and robotic manipulation into the design of fully functional gaze-controlled robotic assistance systems. In particular, it is necessary to develop an overall approach that includes software-only gaze direction estimation, region-based mapping of space, real-time control of robotic arms, and object selection for assistive applications.

This dissertation seeks to address these shortcomings by developing a gaze-controlled robotic assistance system using vision-based gaze estimation and object selection by regions, together with robotic manipulation tasks. In doing so, this study seeks to demonstrate the use of gaze control in human-robot collaboration as an important aspect of making assistive robots more affordable and intelligent.

1. 3 Research Problem

Human–robot collaboration has been steadily gaining traction as a viable approach for designing intelligent assistive systems that help people with physical impairments in carrying out daily activities with greater independence. One of the critical activities that are involved in this process is selecting an object of interest and then retrieving it. It is especially relevant for those people who suffer from serious motor impairments and thus cannot interact with traditional interfaces including keyboard, touchscreen, gesture recognition, and voice commands. Despite the obvious success of the collaborative robotic arm in this area, creating a natural control method for them is still a challenge.

Typically, the methods that are used in order to provide control over assistive robotic arms include mechanical switches, joysticks, verbal commands, brain-computer interface, or special eye-trackers. However, despite their obvious effectiveness in laboratory settings,

they have numerous drawbacks including high cost, difficult setup and calibration procedures, as well as low accessibility. For instance, while commercial infrared eye trackers or special gaze-tracking software may work perfectly well in certain circumstances, they are typically costly, difficult to calibrate, and may require controlled light conditions or even some special hardware.

This raises a fundamental question: How can users, especially users with disabilities who have very severe motor impairments, communicate with robots naturally and autonomously through an inexpensive and non-invasive interface? Tackling this issue will involve looking for alternatives for human-robot interaction that are both effective and easily implemented through widely available equipment.

An interesting idea for achieving this is the use of gaze-based communication. The gaze of a person effectively communicates where he/she is looking at any given time and what his/her intention is. Hence, it becomes a natural way for hands-free communication with a robot for users who cannot utilize traditional control interfaces due to their disabilities. But the process of decoding the gaze of the user into appropriate actions for the robot poses certain difficulties that need to be solved. Thus, the key research problem explored in this dissertation is as follows:

How to design and integrate a cost-effective software-only gaze estimation approach to allow disabled individuals to use a collaborative robot arm for selecting objects via gaze direction?

Notably, this problem does not only involve designing a specific gaze estimation model but poses the more general question of how to build an end-to-end human-assistive human-robot collaboration system that will enable users to translate their intentions expressed in gaze direction into actions of the robot.

One of the key aspects of the problem considered is the development of the gaze estimation algorithm itself. As opposed to hardware-based eye-tracking devices that employ infrared lights and sensors to estimate gaze direction, a software-based approach

requires the computation of gaze direction based only on visual information captured by cameras. Although less costly than hardware-based eye tracking devices, this approach faces a number of difficulties associated with its robustness to changes in environmental lighting, head pose variability, occlusions, inter-user variability, and computational efficiency.

Another challenging issue lies in translating the user's gaze into spatial commands that will enable the robot to manipulate an object appropriately. Gaze doesn't indicate anything about how a robot should move or grasp something. On the contrary, gaze refers to the intention of the user which must be translated into specific spatial movements of the robot. This entails the determination of specific spatial areas related to the objects present in a particular shared environment, prediction of gaze into those areas, and subsequent translation of gaze to robotic actions.

The issue of intention recognition becomes further complicated when it comes to human–robot collaboration because apart from successfully predicting intention, the robot should react appropriately to that intention by performing certain tasks. Specifically, the robot needs to detect the object selected by the user, design a movement pattern, and perform the required task.

An additional problem of research relates to accessibility and inclusiveness. Currently, robotic systems that require multimodal input, including voice, gestural commands, or manual confirmations, are prevalent. For severely impaired individuals, however, these requirements might not be fulfilled. Thus, an ideal system for gaze-based assistance should minimize the use of other modalities while preserving good performance. The question then arises whether eye movements could be sufficient as a tool for robotic control.

A final challenge in relation to research on this topic includes finding the proper balance between cost-efficiency and performance. Precise systems require costly hardware, as well as complex software, while low-budget implementations may lack precision.

Therefore, the problem of developing gaze-based systems for robotic assistance that are both inexpensive yet precise remains unsolved.

In addition, studies in the past have typically regarded gaze estimation, object detection, region mapping, and robotic manipulation as different issues. In assistive human-robot cooperation, however, these different aspects are meant to work together as part of one system. This may lead to complications such as inconsistencies or errors that could reduce usability. Thus, an important aspect of the research question explored in this project is how these different parts can be integrated into one complete system.

Another issue related to this problem is how to manage any possible ambiguities with respect to gaze intention. The gaze could be erroneous, quick, or uncertain. Users may also be gazing about in the environment without the intention of issuing a command. It is thus important to understand how this could be managed in order to avoid mistakes by the robot.

With respect to assistive technology, the problem involves not just technical issues of effectiveness but those related to autonomy and empowerment. Much assistive technology is aimed at compensating for impairments using complex interfaces and does not provide for intuitive human-computer interactions. With its promise of intuitive human-machine interaction, a gaze-controlled robotic assistant presents an interesting alternative, though meeting the promise requires overcoming several obstacles, including perception, intention understanding, robot control, and accessibility.

This dissertation explores this topic in the context of a vision-based approach that uses software-only estimation of gaze direction, mapping of regions of interest, and joint control of a robotic manipulator for selecting objects by means of gaze. Instead of treating gaze estimation purely as a classification problem, this dissertation treats it as part of a larger problem of providing robotic assistance based on human gaze direction.

In particular, this research aims to solve the problem of gaze capture using vision-based techniques with reduced cost, translating it into spatial intent, mapping it to objects in the

workspace, and utilizing it to direct robot manipulator picking actions in real-time through a collaborative approach. Through the investigation of the problem as a systems issue, this dissertation makes contributions toward the design of user-friendly and economical assistive robots capable of enhancing independence among people with disabilities.

More generally, the research problem may be considered as an attempt to bridge the divide between human intention and robotic assistance: how might natural gaze behaviors be translated into an efficient control signal allowing for intuitive, inexpensive, and practical collaboration with a robot?

1. 4 Research Objectives

It is essential to establish that the main aim of this research paper would be the development of the gaze direction-based human-robot interaction framework that will allow differently-abled individuals to select objects by means of gaze direction. This research aims to explore how vision-based gaze direction estimation can be used as an intuitive method to perform robotic manipulation in assistive settings. The purpose of this research is to develop a cost-effective solution for assistive robotic manipulation, which will combine the capabilities of gaze perception, intention recognition, and robotic arm control to turn the users' gaze direction into robotic action.

In addition, one of the important objectives of the research is to design a robust gaze direction estimation model that will enable detection and classification of the user's gaze direction into a number of predefined spatial areas based on RGB cameras. In particular, it is vital to note that the main emphasis of this research will be made on exploration of machine learning and computer vision technologies that will help to perform gaze direction estimation. At the same time, the research will pay special attention to the development of an effective vision-based gaze direction estimation model.

In addition to the above-stated purpose, another important goal of the investigation is to devise a mapping approach for the relationship between detected gaze areas and respective object positions in a collaborative robot environment. In order for the suggested robot to provide a natural choice of objects based on the gaze estimation, there should be a method

for translating the gaze estimation into spatial commands for the robot to act. Hence, the goal of this research is to develop a region-based interaction model in which gaze direction serves as an intention indicator related to target objects in certain workspace areas.

Another goal is to incorporate the developed gaze estimation and region-mapping method into a robotic manipulator for gaze-based object grasping. Although many previous works have focused on the gaze estimation task as a perceptual problem, this work goes further by extending its application into real-world robotic manipulation. The proposed research aims to design a complete pipeline where the robot manipulator is able to recognize the object of interest from the user's gaze direction, get closer to it, and finally grasp it. The purpose of such an integration is to illustrate the ability of using gaze as a means of direct robot control in collaborative environments where typical input mechanisms cannot be used.

Another goal of the research is the assessment of the potential of software-based gaze guidance to achieve effective assistive human-robot cooperation. It is not enough just to develop the framework without verifying its accuracy, reliability, and practicality in actual circumstances. Thus, this research intends to test the developed system with respect to its ability to classify the user's gaze correctly, predict regions of interest, select appropriate objects for manipulation, and demonstrate good responsiveness. Apart from evaluating the efficiency of the proposed approach, this research will also consider the possibility of the framework being user-friendly for people with different capabilities.

A further important goal would be to make a contribution towards low-cost and accessible assistive robotics, through removing some of the barriers that come along with traditional gaze control-based assistive systems. Current gaze-based assistive technologies rely on relatively complex and expensive technology, limiting their widespread adoption. This research project intends to show that it is possible to develop vision-based technologies without the use of specialized hardware components, while still retaining effectiveness and usability. Through such goals, this research project seeks to contribute towards democratization of assistive robotic technologies.

Furthermore, this research will attempt to explore the possibility of gaze-based collaboration between humans and robots, where gaze control can act not only as a passive indication but a method of command for controlling the behavior of robotic systems. Apart from its use in pointing at objects, gaze control is also full of information about the user's intentions. The research thus intends to explore ways in which gaze can be effectively employed as an intuitive means of communication between humans and robots.

In addition, the present work intends to make academic contributions to the related areas of gaze estimation, assistive robotics, and human-robot collaboration by formulating a general framework that incorporates these aspects together. Although many previous studies have focused on these disciplines independently, there is a scarcity of literature where these concepts are considered in combination with software-based gaze estimation and robotic object manipulation toward an assistive application.

Thus, the study shall be based on the achievement of the following objectives:

1. To create a gaze direction estimation model that would operate entirely using software-based techniques based on computer vision and deep learning.

2. To classify gaze directions according to pre-set spatial regions based on user intention.

3. To create the model of mapping between gaze regions and object locations within a robotic work environment.

4. To create the model of human-robot interaction based on the combination of intention recognition and robotic arm control.

5. To test the viability of the designed system.

All of these objectives form the basis for the creation of an innovative system that would allow for gaze direction to serve as a communicational interface in human-robot interactions.

2. Methodology

This chapter describes the methodology applied in designing and evaluating the software-only gaze guidance-enabled human–robot collaboration assistive object selection system. The development of the methodology was guided by the main objective of translating user gaze directions to the corresponding robotic picking actions within a perception and manipulation system. The proposed methodology integrates computer vision-based gaze estimation, mapping of spatial regions, and controlling robotic arm movement such that differently abled users can perform object selection by just directing their gaze.

The proposed methodology diverges from common practices that rely on specialized gaze recognition hardware or multiple sensors to achieve effective assistive human-robotic collaboration. Instead, it relies on inexpensive imaging devices and deep learning techniques to estimate gaze direction. The methodology provides means for effective human-robot collaboration without compromising the accuracy of gaze detection and robotic action.

The methodology employed in this study includes a number of steps. They include dataset creation, gaze-region modeling, image pre-processing, deep learning-based gaze estimation, mapping of intention based on gaze region, object selection using robotic arm, and system evaluation.

2. 1 Research Design

This research uses an experimental design research methodology. This entails that there is integration of system development with empirical performance testing. The prototype method is used, whereby a comprehensive framework is developed, deployed, and tested through repeated experiments.

The methodology is structured around three major components:

- Vision-Based Gaze Estimation Module
- Gaze-to-Region Mapping Module

- Robotic Manipulation Module

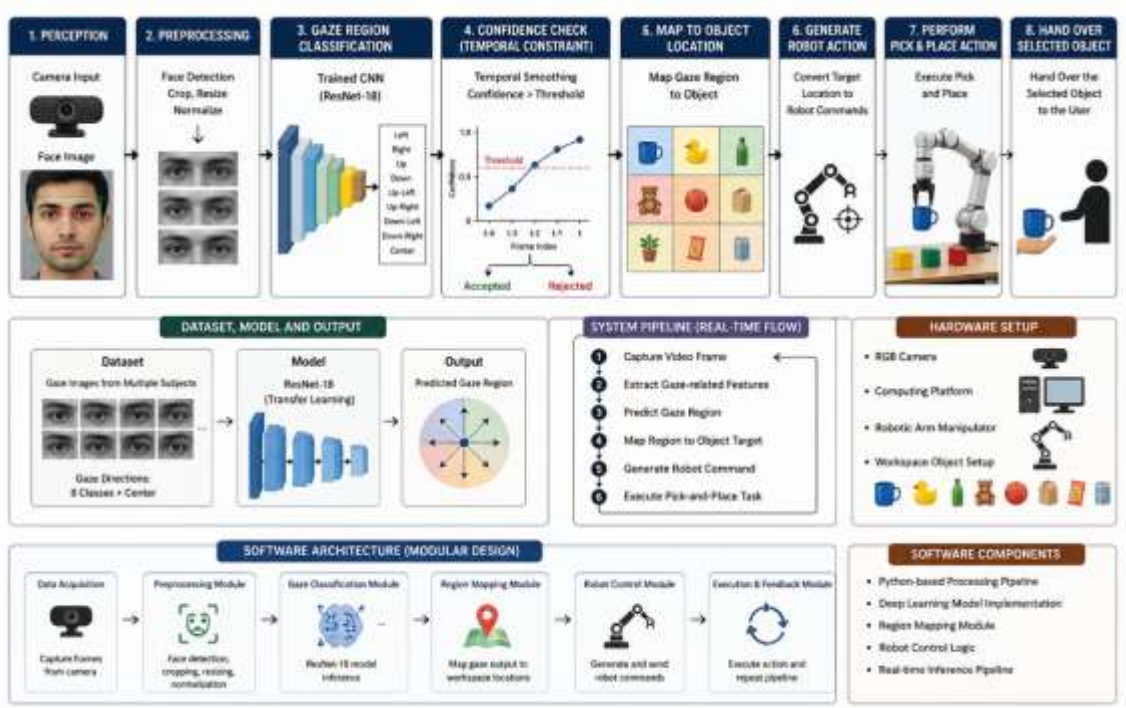

Figure 1. Research Design

The suggested framework comprises three connected functional modules which work together to enable gaze intention recognition and subsequently robotic task completion based on the analysis of gaze direction performed by the user. The first module deals with gaze perception during which user's gaze direction is computed based on computer vision and deep learning technologies used to estimate gaze direction based on purely software-based vision models. This module processes visual information related to user's facial and gaze features collected via RGB cameras and determines user's gaze in pre-defined regions inside the workspace of a robotic agent.

The second module involves intention interpretation based on region-based mapping. During this process, user's gaze regions will be considered as signs of their intentions and mapped into target regions or objects in the workspace. As a result, the system will be able

to interpret intentions of the user expressed as gaze regions, which can be seen as selection instructions. Thus, the transformation of user's gaze direction into selections will allow establishing a link between perception and robotic control modules.

The third module will be in charge of performing the manipulation and tasks for robots. According to the selection of the targeted region, the robot arm locates the particular objects and performs the necessary picking up and placing down process. This comprises the processes of creating target coordinates, motion planning, object grasping, and object retrieval. By accomplishing the third module, the whole gaze-based human-robot collaboration system will convert visual intention of users to robotic actions physically. The above three modules constitute the complete gaze-based collaboration system.

The complete system operates as a human-intention-to-action pipeline:

User Gaze → Gaze Classification → Region Mapping → Object Selection → Robot Pick Action

The hierarchical structure facilitates the design, testing, and optimization of the individual modules within this framework independently but as a part of the integrated collaboration system. The framework makes it possible for the developers to design and optimize each of the components independently, and their interactions result in the creation of the whole framework as a working collaborative system. At the same time, this approach to the organization of the components contributes to seamless information transfer from gaze detection to robotic action. As a result, the proposed architecture makes it possible for the system to operate as an integral human-robot collaborative system capable of translating human intentions into robot actions.

2. 1 Proposed System Architecture

The architecture of the proposed system involves several stages of processing in order to convert the intention of the user gaze into the desired assistive action from the robotic arm via the integration of perceptual, decision-making, and action phases. At the first stage of this architecture, visual perception takes place. In particular, the facial area including the

region around the eyes of the user is captured by an RGB camera. Next, the captured images are analyzed via a gaze estimation pipeline including preprocessing of the image, facial feature detection, and detection of gaze cues like the direction of the eyes and their positions. Instead of estimating the continuous coordinates of the gaze, the proposed system implements a region-based gaze classification method that assigns the user gaze to pre-defined regions.

This output from the classified gaze then proceeds to be fed into an intention-mapping module, where predicted gaze areas are mapped onto object targets as well as workspace coordinates. This is essentially the decision-making phase for the proposed architecture. After identifying the target area of the gaze, the coordinates that have been mapped can be sent on to the robotic manipulator, which will carry out the desired picking and placing of the object based on motion planning and grasping. As such, this architecture is able to create an end-to-end solution that combines vision and intention recognition as well as autonomous robotic reaction to these inputs.

The architecture includes:

- Image acquisition module
- Face and eye preprocessing module
- CNN-based gaze classifier
- Region mapping module
- Robot control module
- Feedback and validation module

The advantage of this architecture is that it promotes scalability and ease of integration between perception and robotic action, with greater flexibility, maintainability, subsystem optimization, and the ability to expand the system in the future.

Dataset Development

For this study, a customized gaze direction database was created to develop a gaze classification model. As existing databases generally concentrate on gaze estimation and eye tracking applications, a specialized database was constructed for this particular application.

The seven classes of gaze directions, which corresponded to workspace regions, were:

- Upper Left
- Upper Center
- Upper Right
- Lower Left
- Lower Center
- Lower Right
- Ambiguous/Neutral

These regions indicate where objects might be located within the workspace of the robot and serve as the foundation for the gaze-based selection approach.

The process of data gathering included recording the faces of the subjects while they were focusing their gaze on target markers in different spatial locations.

Variations included:

- Lighting conditions
- Head pose differences
- Eye orientations
- Facial appearance diversity
- Camera angle changes

The above variation was purposely included to enhance the ability of the proposed gaze estimation model to generalize in practical operation scenarios by being more robust towards changes in factors such as illumination, face appearance, head pose, and camera views. Including the above variations was aimed at simulating practical interaction scenarios that would minimize overfitting in practical environments where such variations occur. The recorded samples were then classified manually into folders corresponding to different gaze areas for effective and consistent training using supervised learning techniques.

Dataset structure:

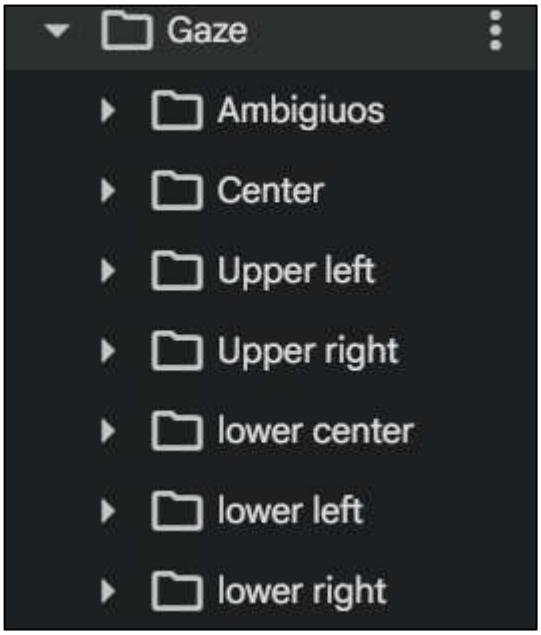

Figure 2. Data structure

The aforementioned well-structured database served as the basis for training and validating the proposed eye movement prediction technique, where it helped with supervised training, performance evaluation, regional classification, and validation. Data Preprocessing

Before proceeding to train the proposed model, a thorough preprocessing procedure was conducted with the aim of improving data homogeneity, increasing the quality of images,

and eliminating the presence of unnecessary variance from the dataset which would hamper model performance. This preprocessing was critical for preparing input samples so as to make them appropriate for deep learning feature extraction. Specifically, the input images were scaled to certain dimensions according to the architecture used for CNN-based learning, thus making sure that input samples were presented uniformly. Pixel normalization was also performed to adjust image intensity values to the same scale, helping optimize the whole learning procedure by enabling fast convergence of the algorithm.

Preprocessing steps included:

- Image resizing
- Pixel normalization
- Face-region alignment
- Noise reduction
- Data balancing

Generalization and Overfitting Reduction Techniques were used for enhancing generalization and avoiding overfitting, techniques such as data augmentation were implemented in order to introduce greater variability to the training process and make it more realistic.

Augmentation operations included:

- Rotation
- Horizontal shifts
- Brightness variation
- Minor zoom transformations
- Synthetic perturbations

These transformations allowed the simulation of realistic variations while maintaining the accuracy of gaze-labels. The preprocessing pipeline helped achieve consistent representation, feature learning, model robustness, and reliable gaze-classification results.

Gaze Direction Classification Model

Gaze direction classifier, which is based on deep learning techniques, is considered to be the most important part of the proposed perception module. This particular component was created specifically to detect users' gaze direction on the basis of an image of the face and to ensure gaze-driven human-robot interaction. As it was already mentioned, correct detection of gaze intention is crucial for the proposed framework.

The proposed approach uses CNNs as they have been extensively used for image classification and extracting visual features from images. CNN models are suitable for gaze estimation as they can extract hierarchical features of the visual patterns corresponding to eye orientation, face geometry, and other appearance cues without having to create custom features manually. The lower layers of the network usually extract visual primitives like edges, lines, and textures, whereas the higher layers extract higher-order semantic information about the gaze direction and head orientation. Thus, CNNs are suitable for gaze classification at different regions.

Out of various deep learning models studied, ResNet-18 is chosen as the base classification model owing to the model's good trade-off between high classification accuracy, complexity, and efficiency. ResNet-18 is an instance of residual network architecture with 18 trainable layers that make use of residual connections for better feature extraction in deep learning.

One of the prominent features of ResNet-18 is the utilization of residual learning technique. In residual learning, shortcut connections allow bypassing certain layers by transmitting the information through them, which allows overcoming problems associated with vanishing gradient problem in deep neural networks. Instead of having individual

layers learn complete transforms, residual connections allow the network to learn residual mapping functions.

This residual learning principle can be expressed as:

$$\boxed{H(x) = F(x) + x}$$

where:

H(x) = desired mapping
F(x) = residual function learned by the network
x = input feature representation

This structure helps the network retain valuable features even as it learns deeper representations pertaining to gaze. The network was trained for classification of gaze into seven different directions corresponding to gaze regions:

- Upper Left
- Upper Center
- Upper Right
- Lower Left
- Lower Center
- Lower Right
- Ambiguous/Ne

The gaze classification formula converts the problem of estimating gaze direction into one of supervised multi-class image classification.

For input, we have pre-processed images of the face or eyes obtained from conventional RGB cameras, and for output, we have gaze region classifications.

The classification function can be represented as:

$$\boxed{f(x) \rightarrow \text{and}_i, i \in \{1, \ldots, 7\}}$$

where:

x = input image

yi = predicted gaze region

The mapping between input images and the gaze region labels is learned by the network using supervised training, which employs the labeled dataset created for this project.

The training of the ResNet-18 model comprised several optimization methods that sought to improve classification accuracy while minimizing overfitting. The cross-entropy loss function was utilized for the multi-class classification problem, whereas backpropagation was used for optimization.

- Cross-entropy loss optimization
- Backpropagation
- Validation monitoring
- Early stopping
- Hyperparameter tuning

The cross-entropy function was utilized as a measure to evaluate the penalty for classification errors while validation was employed as a tool to assess generalization abilities in the training process. Early stopping was also included in order to avoid overfitting.

Hyperparameter tuning was performed to optimize parameters such as:

- Learning rate
- Batch you
- Number of epochs

- Optimization strategy

These tuning steps contributed to improved model convergence and classification robustness.

Another criterion that made ResNet-18 ideal for use in the project was its ability to strike a balance between classification accuracy and computational efficiency. Relative to the deeper ResNet models like ResNet-50 and ResNet-101, ResNet-18 is computationally efficient yet capable of performing well in terms of classification accuracy.

The effectiveness of choosing ResNet-18 over other CNN structures such as MobileNet and VGG16 was verified by comparing their performances.

One of the factors that made MobileNet a good choice was its high efficiency. Despite being computationally efficient, it had poor discrimination ability in this case.

As demonstrated through comparative experiments, ResNet-18 performed better in terms of overall classification ability, higher level of regional discrimination, and greater generalization across different gaze classes and is therefore the chosen classifier for deployment.

In addition to accuracy of classification, the selected neural network was also found to be resilient against changes in lighting and appearance of the face, thus being suitable for use in gaze-based assistive tasks.

In summary, the deep neural network gaze classifier based on ResNet-18 represents the basis of perceptual system that can be employed for gaze direction estimation. In turn, it will form the basis for region mapping and robotic selection of objects of interest.

Region-Based Intention Mapping

After identifying the gaze direction, the predicted output should be interpreted as user intention to facilitate collaborative robotic actions. Unlike considering gaze as a mere observational signal, the current study makes use of gaze as a command signal that allows the conversion of user attention into a decision on the object to select. Under this

consideration, each gaze category corresponds to an object-selection area in the robotic work area, which facilitates the use of gaze direction as a natural mode of control for human-robot collaboration.

The approach of using regions for the interaction scheme involved matching the categorized gaze direction with the target objects located at the corresponding predefined workspace areas. For instance, gaze directed toward the upper left, center, and right workspace areas will be considered as different object targets. Similarly, the same can be done for lower gaze directions if relevant target areas exist.

To facilitate this task, a specialized region mapping module was introduced in order to map the predicted gaze regions to their respective workspace coordinates for the robotic manipulator. After the region of interest is determined through gaze prediction, it will be mapped to its corresponding target location on the robot workspace which will then be used to generate the required command instructions for the robot. This way, the prediction of gaze regions provides the high-level information about the intent of the user, which will then be translated into action through the mapping module.

It can be illustrated through the following equation:

Region Prediction → Workspace Coordinate → Robot Command

The above-mentioned process establishes the crucial link between gaze prediction and robotic manipulation. Not only does it provide an opportunity for the proposed framework to go from basic gaze recognition to more advanced collaboration based on visual intentions but also simplifies the control mechanism, minimizes confusion in regards to selecting the desired target and ultimately enhances the feasibility of utilizing such gaze prediction-based interactions for assisting differently-abled individuals.

Object Selection Logic

Once the intention mapping is done, the selection process is determined by the object selection rationale.

This rationale depends on:

- Gaze region prediction
- Object-region association
- Thresholds of confidence
- Conformation of stability

Consistency check techniques have been implemented over time to reduce errors arising from gaze noise. Object selection can take place only if:

- Consistent region prediction is achieved in a specific number of frames
- Confidence level of predictions is greater than a specific value

Thus, the performance of the assistive interface is greatly improved.

Then, the target coordinates will be delivered to the robotic arm controller via the object selection module.

This chapter discussed the approach employed in the study by explaining research design, system architecture, datasets generation, data preprocessing, gaze classification, region prediction, and object selection.

Robotic Arm Control Architecture

After estimating gaze direction and mapping out intentions, the third key part of the suggested framework involves the robotic arm controller tasked with transforming gaze intentions into manipulation of the object in question. This component acts as a bridge between the user's intention and execution of a robotic task within the proposed human–robot interaction paradigm.

The robotic manipulator used for this experiment was created in order to function as an assistive pick-and-place robot that will be able to move an object related to the gaze region

detected by the software. When the target region is determined using the gaze prediction and mapping techniques, spatial coordinates are transmitted to the robot arm.

The architecture of the robot control mechanism includes a number of interlinked layers such as:

- Target coordinates creation
- Motion planning
- End effector manipulation
- Grasping object
- Picking and placing sequences

These layers contribute to the process of picking up objects independently in accordance with the intention inferred from the gaze point.

Initially, the gaze classification component verifies whether there is an adequate target region, which is subsequently translated into the corresponding object coordinates in the workspace. Then, the robot control mechanism develops motion directions for the robotic manipulator.

The pick-and-place sequence of events for the robot involves a pipeline process as follows:

Gaze Selection → Target Coordinate Recognition → Robot Movement Planning → Object Grasp → Object Retrieval

Such an arrangement enables user gaze to act as a means of issuing high-level commands, whereas the robotic system handles low-level movements.

To accommodate the possibility of using multiple robots in future projects, a modular control design was considered necessary.

Workspace Region Mapping for Robotic Picking

For the purpose of ensuring proper responses from the robot, a structured environment model was created where objects were arranged in a systematic way within pre-specified areas that corresponded to the specified gaze directions. Such structuring of the environment created a link between the intention of the user's gaze and the location of the object, making it easier to make proper selection decisions for the robot. In order to do so, the environment was partitioned into multiple selection areas that corresponded to gaze direction classification areas, thereby enabling mapping of gaze output predictions to object targets.

Example region configuration:

Gaze Direction	Mapped Object Region	Description
Upper Left	Region 1 (Left)	Selects object positioned in left workspace region
Lower Left	Region 1 (Left)	Selects object positioned in left workspace region
Upper Center	Region 2 (Center)	Selects object positioned in center workspace region
Lower Center	Region 2 (Center)	Selects object positioned in center workspace region
Upper Right	Region 3 (Right)	Selects object positioned in right workspace region
Lower Right	Region 3 (Right)	Selects object positioned in right workspace region

Table 1.Region Configuration

The following were assigned to each of these regions:

- Identity of the object
- Cartesian coordinate of the object target
- Position required to grab the object
- Path for approaching safely

Such a mapping establishes a clear correlation between the user's intent and the target of the robot.

The spatial mapping system can be defined as follows:

Gaze Region → Target Coordinate (x, y, z)

The prediction made will then be converted to the workspace coordinate needed for robotic movements.

Motion Planning and Pick-and-Place Execution

After the identification of the target object's coordinates, the robot motion planning involves the generation of trajectories that facilitate a safe and efficient manipulation process of retrieving the object without any mishaps.

Motion planning was intended for:

• Collision avoidance

• Approach path for safety

• Grasp positioning

• Controlled transportation

The pick-and-place operation by the robot involves several phases of motion:

1. Move to pre-grasp pose

2. Alignment of end effector with target object

3. Execute grasp

4. Lift object

5. Transport object to desired location

6. Release object

These form the steps involved in the pick-and-place operation cycle.

The motion control strategy was:

Target acquisition → pre-grasp pose → grasp execution → delivery pose

Generation of motion trajectories involved minimizing sudden and erratic movements while ensuring accuracy.

The grasp execution phase entailed the alignment of the end effector relative to the position of the object. The grasp could be executed only when the target was localized.

Object Detection Support for Robotic Selection

For the purpose of improving localization of the objects to be picked up, an object detection layer was added to help with the robotic manipulation task through additional sensing of the environment and confirmation of the target object. Although gaze estimation helps in determining which area on the workpiece the user desires, the object detection process aids in localizing the specific target within that selected area for accurate determination of its position. This study uses vision-based object localization as an additional layer to gaze-based selection.

Object detection supports:

- Target confirmation
- Object boundary localization
- Grasp point estimation
- Error reduction

Deep learning object detection approaches such as:

- YOLO
- CNN-based detection methods

can support this layer.

The incorporation of eye-tracking-based intention with object perception increases the robustness of robotic grasping by incorporating user intentions along with their environment perceptions to accomplish tasks with increased reliability. The approach

establishes a double perception paradigm, wherein intentions from eye tracking and object perception through detection are combined to guide robotic action. Through intention and environment perception together, increased target selection precision is accomplished, while the interaction between human and robot becomes more efficient.

System Implementation

The designed framework was developed into a software and hardware combined pipeline which had perception, decision-making, and robotics execution units, all working as one integrated system. In terms of hardware components, there were an RGB camera as an input data capture device, computer for real-time data processing and deep learning model inference, robotic arm manipulator as a picker, and workspace environment with objects in predefined regions. The software part comprised the processing pipeline written in Python, the implementation of the gaze classification model based on deep learning, region interpretation, robot control code for moving actions, and finally, the inference pipeline for real-time interactions. The workflow followed a series of steps where frames from videos were captured, gaze relevant features extracted, gaze region was classified, mapped to an object of interest, robot commands for pick-and-place operation were created, and actions were executed. This process kept repeating, providing the opportunity for continuous assistive control of the process based on the user's eye-tracking. Finally, each component of the designed architecture was implemented independently so that testing and optimization could be done separately.

2.3 System Algorithm

The overall system logic can be summarized through the following algorithm.

Algorithm: Gaze-Guided Object Selection

Step	Process	Description
1	Image Capture	Gain facial or image frames using RGB/CCTV cameras for the gaze analysis task.

2	Preprocessing of Facial & Gaze Input	Crop and preprocess the facial input using face detection algorithm, resize, normalize, etc., before providing it to the CNN classifier.
3	Gaze Region Classification Using Trained CNN Classifier	Using ResNet-18 pre-trained model classify the predicted gaze region of the user.
4	Check Prediction Confidence	Check prediction confidence by applying temporal constraints between frames and check if the result is above threshold value.
5	Map Predicted Gaze to Corresponding Object Location	Given that certain objects lie in certain gaze regions, map the classified gaze to the relevant object locations.
6	Generate Robot Action	Translate target location to robotic arm movements and commands.
7	Perform Pick and Place Action	Instruct robot to perform pick and place action based on the robot action generated in the previous step.
8	Hand Over Selected Object	Handing over the object picked by robotic arm to human

Table 2.Gaze guided object detection

The entire procedure is repeated recursively until the termination of user input engagement, thereby ensuring a seamless human-robot interaction process. The whole system works under the decision-making control paradigm where any gaze prediction obtained is always first validated before any robotic task initiation. Should the algorithm detect an acceptable gaze prediction with sufficient confidence level, the predicted gaze location is then considered to be the target object selected by the user. It is then followed by the robotic manipulation task execution where the robot picks up the selected object from its workspace and moves it to the desired target location. However, should there be no sufficient confidence level for the predicted gaze prediction or it was found to be unstable, the algorithm will not initiate any command for object selection. The process then repeats again until the gaze prediction is stable enough and can be used to make decisions.

2.4 Experimental Setup

Both gaze estimation performance and gaze-guided robot object selection was tested through experiments.

The setup for experiments included:

- Gaze data collection-controlled environment
- Labeled region object workspace
- Robot arm assembly
- Real-time inference environment

Experiments focused on assessing both perception level and task level performance.

Experiments were performed using two different types:

1. Gaze Classification Experiment
2. Robot Object Selection Experiment

Gaze classification experiment involved evaluation of CNN models in terms of prediction performance.

Experiment on robotic aspect focused on determining whether gaze guidance can successfully lead to object selection.

Several rounds of experiments were run, taking into account various conditions, including:

- Variations in lighting
- Variation in head position
- Variations in object placement
- Variation in selection task repeated performances.

2.5 Performance Evaluation Metrics

System performance was evaluated using standard classification and task-performance metrics.

For gaze estimation:

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Additional metrics:

- Precision
- Recall
- F1-score

Confusion matrix analysis was conducted as part of the model evaluation approach to determine region-based prediction accuracy and reveal classification characteristics associated with the established categories of gaze direction. Confusion matrix analysis helped determine the extent to which the proposed model could effectively differentiate between regions of gaze direction while pointing out possible misclassification issues, class overlaps, and ambiguities. By evaluating true and false predictions per region, confusion matrix analysis offered an effective method for analyzing discriminative abilities at the level of class prediction, determining patterns of error, and interpreting results.

Apart from gaze classification accuracy, the proposed model framework was also intended to be evaluated using several task-based metrics that could help measure the feasibility of using gaze direction intentions for object manipulation and selection. One evaluation criterion considered was object selection success, referring to whether the robotic system correctly identified and moved toward the object corresponding to the user's intended gaze region during testing scenarios. This criterion was used qualitatively to observe whether gaze predictions could successfully guide target selection.

One more consideration was the ability of the robot to perform the entire process of picking up an object and delivering it. This was taken into account while checking the performance of the whole system.

Responsiveness of the system was one more implementation criterion that was considered important to be met. Instead of measuring latencies as part of formal benchmarking tests, the responsiveness of the system was checked via its behavior.

Finally, the reliability of the tasks performed was also considered. This involved the accuracy of eye tracking for object identification as well as the actions performed by the robot for each individual instance. These measures of evaluation were more concerned with testing the practicality and feasibility of the proposed collaborative human-robot assistance framework.

2.5 Safety and Reliability Considerations

Since the framework works in collaborative human-robot settings, safety and reliability aspects were considered during development.

Safety factors consist of:

- Threshold confidence prior to taking action
- Multiple frame gaze verification
- Limited robot movement
- Fail-safe stopping criteria

These will minimize false actions resulting from inaccurate predictions of gaze points.

Reliability was improved by:

- Region stability validation
- Error management strategies
- Separate system component testing

This is especially vital when designing assistive technologies for differently-abled individuals.

This chapter covered the robotic manipulation and implementation process for the suggested eye-gaze-controlled assistance framework. This chapter discussed robotic arm control structure, workspace modeling, motion planning, implementation process, algorithm, experiment design, evaluation criteria, and safety measures. Along with the perception process detailed in the previous section, all of these processes make up the entire methodology for converting eye-gaze intention into collaborative robotic object selection.

2.6 Commercialization Aspects of the Product

The suggested software-based human-robot collaboration system utilizing gaze input can be commercialized effectively as an assistive technology solution in the healthcare, rehabilitation, educational, and intelligent robotic markets. While many traditional gaze control technologies require costly dedicated eye tracking equipment, the suggested solution is based on inexpensive computer vision and robotic components, making the system commercially attractive. The ability of the system to interpret gaze intention and use it to control robotic object manipulation opens commercialization avenues in terms of both an autonomous assistive technology product and intelligent interaction software.

Among the main commercial advantages of the suggested system is that it can be marketed as an assistive device, as the proposed solution can provide a cost-effective solution for people suffering from motor disabilities. Traditionally, assistive robotic solutions include costly proprietary equipment, which makes their widespread adoption challenging in developing countries where many users cannot afford to purchase such expensive robotic devices. Utilizing cheap computer vision techniques for gaze intention recognition and robotic manipulation can create opportunities for commercialization as an assistive device aimed at differently-abled users.

A possible application of the product from this study could be a gaze-controlled assistive robotic workstation that would enable users to pick up everyday objects by gaze alone, thus improving their ability to complete daily tasks autonomously. The technology described here might be commercialized for use in residential settings, rehabilitating

institutions, facilities for people in assisted living conditions, and special education institutions. Potential application areas include medication acquisition, classroom aid, object passing, and other tasks requiring the assistance of a robotic workstation for physically impaired individuals.

Besides applications in assistive technologies, the proposed system also offers prospects for commercialization in a more general area of human–robot cooperation. With collaborative robotics becoming increasingly widespread in various applications, including industry and services, gaze-based intention recognition is expected to prove useful as an interaction method for shared human-robot workspaces. The proposed approach might be repurposed as an additional human intention interface component in the framework of collaborative robots. Thus, the product could be commercialized as a software component or intelligent interaction add-on for robotic platforms.

Scalability of the software is another vital consideration when discussing commercialization. Since gaze estimation is implemented in software, there are more ways to commercialize the product aside from hardware sales. For instance, the technology could be packaged as:

- Product bundle for assistive robotics
- Licensing of the software
- Intelligent control module for embedded use
- SaaS platform for robot interaction

This enhances the chances of commercial success through different business models.

There may be other avenues for commercializing the product through subscriptions or software licensing. Under this scenario, the software can be licensed to organizations like rehab facilities, schools, hospitals, or robotics companies for their respective use cases.

As a product innovation, the proposed solution also provides competitive advantages in terms of its differentiated features that include software-based gaze control, inexpensive

cost, ability to select objects robotically, and focus on accessible human-robot interaction. This approach to designing assistive interfaces can offer some competitive benefits over existing solutions involving physical controls or expensive eye tracking devices for people with motor impairments.

Another feature of the proposed design that allows the potential product to be commercially viable is modularity. At present, the system only performs gaze-guided object picking operations, although other intelligent functionalities may be implemented in commercialized versions, including:

- Multimodal control using voice assistance
- Homes devices interactions
- Wheelchair/mobility assistance
- Personal assistant robots
- Smart adaptive interaction models

Opportunities for commercialization within the target market exist in various industries. For instance, one target market opportunity for commercialization is in assistive healthcare technologies. There has been increasing demand for intelligent systems that would help users perform tasks independently. Moreover, there are opportunities in the field of rehabilitation robotics. Gaze guidance systems would serve as an important tool in assisting patients during therapy.

Additionally, educational support technologies that would assist differently-abled learners could constitute another market for the proposed product. Inclusion and innovation are two growing trends in the realm of technology. Governmental and healthcare institutions are actively investing in technologies that enable independence. Therefore, the proposed product aligns with this trend.

Considering a strategy for commercialization, the first phase of product commercialization from prototype may be a staged development process. The first stage

will entail prototype validation and pilot demonstration in controlled assistive conditions. The second stage can involve product enhancement through user testing and hardware optimization. The third stage may facilitate commercialization through collaborative ventures with robotics manufacturers, assistive technology vendors, or startup incubators.

Commercialization potential may also include intellectual property prospects. New inventions relating to gaze-directed robotic decision algorithms, region-intention mapping technologies, or assistive systems architecture can present patent possibilities or software frameworks. Intellectual property protection can bolster commercial viability in the long run.

A preliminary commercialization business model may incorporate both hardware and software aspects. The product portfolio may consist of:

Core Product:

- Gaze-guided assistive robotic system

Revenue Opportunities:

- Device sales
- Software licensing
- Institutional subscriptions
- Maintenance and support services
- Customized deployment solutions

Such a hybrid business model that's going to support both scalability and sustainability.

Even with the possibilities of commercialization, there are many other practical issues that must be considered before full-scale implementation. Some assistive robotic products may need to be validated concerning safety and accessibility guidelines. All these aspects are part of the overall commercialization process and product development life cycle.

In conclusion, the proposed robot exhibits good commercialization potential by virtue of the technological innovation, assistive function, low-cost structure, and broad applicability to various industries. Through addressing both the market demands and accessibility concerns, the study not only qualifies as an academic model but serves as the foundation for an actual intelligent assistive device product.

In short, the commercialization objective of this study is not limited to the demonstration of a robotic prototype but aspires to develop a platform for intelligent assistive technology innovation.

2.6 Testing and Implementation

The proposed software-based gaze-guided collaboration model between human and robot has substantial commercialization prospects in the realm of assistive technologies products, especially within the domains of healthcare, rehabilitation, education, and intelligent robotics. Unlike most existing gaze-based systems, which depend on specialized and expensive eye tracking devices, the proposed technology utilizes cheap technologies like computer vision and robotics. The potential of converting gaze intention into object manipulation by the robot makes it a viable product for commercialization either as a standalone assistive device or as an intelligent interaction platform.

Commercially, one of the greatest advantages of the proposed system is its capacity to serve as a relatively affordable assistive device due to its technological advantage over most of the existing assistive robotic devices. Existing assistive robots make use of highly expensive proprietary hardware components, limiting their commercialization in low-income communities. By using the simple software-based gaze estimation with inexpensive RGB camera and then using this data to control the movement of robotic hands, the proposed device becomes highly affordable.

One possible commercial application that could result from this research would be the development of a gaze-driven robotic assistive workstation. With such a product, users would have the ability to select and fetch ordinary objects by making use of only gaze.

Thus, such a solution might be commercialized for usage at homes, rehabilitative centers, assisted living facilities, and special educational institutions. Possible applications of such technology would include object fetching, helping in classrooms, object transfer, as well as assisting people who suffer from various disabilities.

On the other hand, such an approach can find applications in human–robot collaboration domain. As collaborative robots continue to enter various industries and become more widespread, there will be a need for a means to effectively interact with them. Gaze-based intention recognition technology might become a useful tool for human interactions with robots working in a collaborative mode. Such a system could be commercialized either as a dedicated robot interface layer or as an add-on intelligent interaction component to current robotic platforms.

One of the potential ways how this kind of research can be applied in commerce is by creating gaze-based assistance robot workplace. In that case, users will be able to grab and retrieve any ordinary object just by using their gaze. Therefore, this kind of technology could be sold and used in homes, rehabilitation centers, assisted living homes, and also educational institutions. Possible applications in this area could include retrieving objects, aiding teachers during lessons, transferring items and also helping disabled people.

Alternatively, this technology can be used in the domain of human–robot collaboration. With increasing implementation of collaborative robots in different areas of production, there is going to be a need for efficient way of communicating with them. Intention recognition based on gaze can become one of the solutions for human interaction with collaborative robots. This invention can be implemented as a separate software interface or as an additional layer of intelligent interaction with currently existing robotic systems.

In terms of product innovation, there are a number of elements in the proposed design that can be considered as providing differentiation for competitive advantage. These are software-based eye tracking, affordable construction, capability for robot-controlled object picking, and an accessibility-oriented Human Robot Interface (HRI). It is believed

that the abovementioned factors may allow for competitive advantage against existing assistive technology solutions that require manual controls or costly eye trackers.

Another aspect that should be noted is related to the possibility of using the proposed technology for product extension. The present application provides a simple example based on eye-controlled object selection; however, further versions intended for commercial release can offer additional capabilities, such as:

- Multimodal voice control
- Control of home appliances
- Integration with wheelchairs and other mobility technologies
- Personal robotics and smart assistants
- Advanced personalized interaction

There are multiple target market opportunities available for commercialization. For example, assistive healthcare technologies is one of the biggest potential markets since there is a growing demand for smart solutions that would help people with their daily routines in order to maintain their independence. Also, rehabilitation robotics may become a market opportunity for gaze-controlled technology because it may serve both as an assistive technology during therapy and as a tool for assisting users.

Furthermore, educational technologies for differently abled students may be considered as well because this market offers numerous opportunities for commercialization. There are more and more opportunities in the area of inclusive technology and socially responsible innovation that may become additional drivers for the commercialization process of the proposed product.

Under the commercialization strategy, the prototyping to product process might be approached in a phased manner. First, there is the prototype validation stage with pilot implementation in assisted settings. Secondly, the product improvement stage would entail user testing and hardware enhancements. Thirdly, the commercialization stage might

involve collaborations with robotics companies, assistive device suppliers, or start-up incubation approaches.

The commercialization strategy might also involve intellectual property considerations. Original innovations in areas like gaze-directed robot selection algorithms, intention region mapping techniques, or the architecture of the entire assistive device system might present potential patentable products or proprietary software frameworks.

A possible business model for commercialization may consist of both hardware and software components. This product line may involve the following:

Core Product:

- A robotic assistance system based on gaze guidance technology

Sources of Revenue:

- Product sales
- Software licenses
- Institutional subscriptions
- Maintenance and support services
- Specific deployment solutions

It is possible that the suggested business model would help achieve scalability and sustainability.

Although there exists significant commercialization potential, issues like reliability, regulatory compliance, user certification, and manufacturing cost have to be considered prior to large-scale implementation. Assistive robotic devices may need to be validated based on their safety, accessibility, and human-centricity requirements. These points contribute to the commercialization process and the product maturation cycle.

Finally, the current project shows high potential for commercialization, given its innovation, assistive value, economic feasibility, and industry application. As a result of combining market demand and accessibility problems, the research serves not just as an academic prototype, but as an innovative intelligent product as well.

In conclusion, the commercialization perspective of the current project includes scaling up the initial prototype into an innovative assistive technology platform.

3. Results and discussion

3.1 Results

In this section, we analyze the results of experiments carried out on the proposed gaze direction estimation software approach for implementing the gaze-guided human–robot collaboration system. Various results such as the dataset statistics analysis, model training analysis, classification analysis, and inference analysis will be discussed.

A variety of analysis approaches were considered to investigate various characteristics of the proposed human–robot collaboration solution, including the dataset preprocessing stage, the training stage of the developed deep learning-based model, the classification accuracy, and the inference efficiency. Some of the results are quantitative in nature, while others involve visualization approaches like confusion matrixes, Precision/Recall, F1-score calculations, accuracy and loss training curves, class distribution graphs, image quality distribution plots, and inference analysis on unseen samples.

Dataset Exploration Results

Additionally, the discussion explains the importance of the results in terms of assistive robots, gaze-based interaction, and collaborative robotic systems.

Exploratory data analysis was performed before modeling to determine the nature of the data and its applicability to the classification process. This combined data set contained facial images that were well balanced in terms of the defined gaze direction categories.

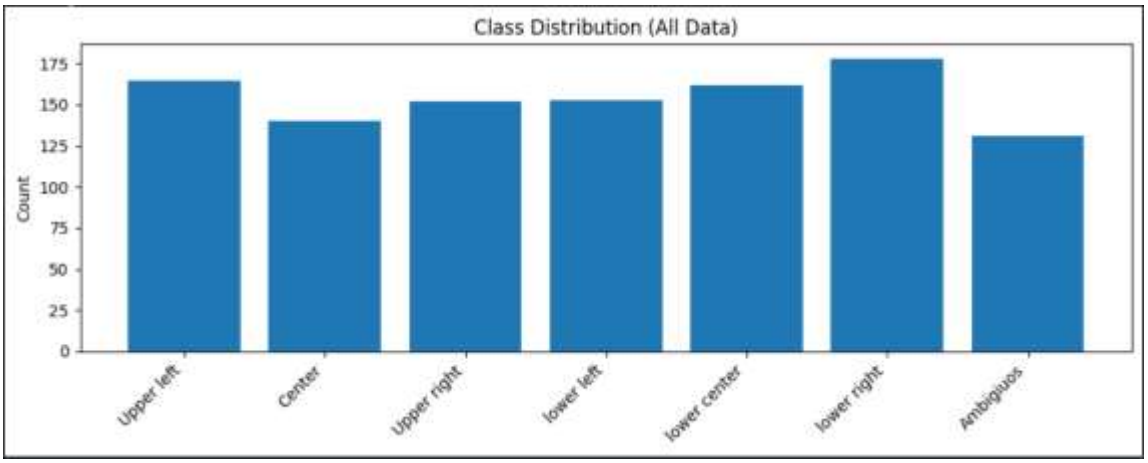

Figure 3. Class-wise Sample Distribution

It is apparent from the above class distribution that there is fairly equal representation for all classes of gaze. This means that there would be no chance of having class imbalance, which might affect the model training. The above class distribution plays an important role in making the model more effective while classifying the data sets.

Further analysis at the level of the entire dataset was carried out to investigate possible features of images which could affect the effectiveness of model training, such as lightness, contrast, sharpness of the images, and dimensional similarity. The examination of such characteristics is useful both for evaluation of the quality of the dataset and for identification of differences in the conditions under which the images were taken, which is necessary for obtaining high-quality training inputs for the model.

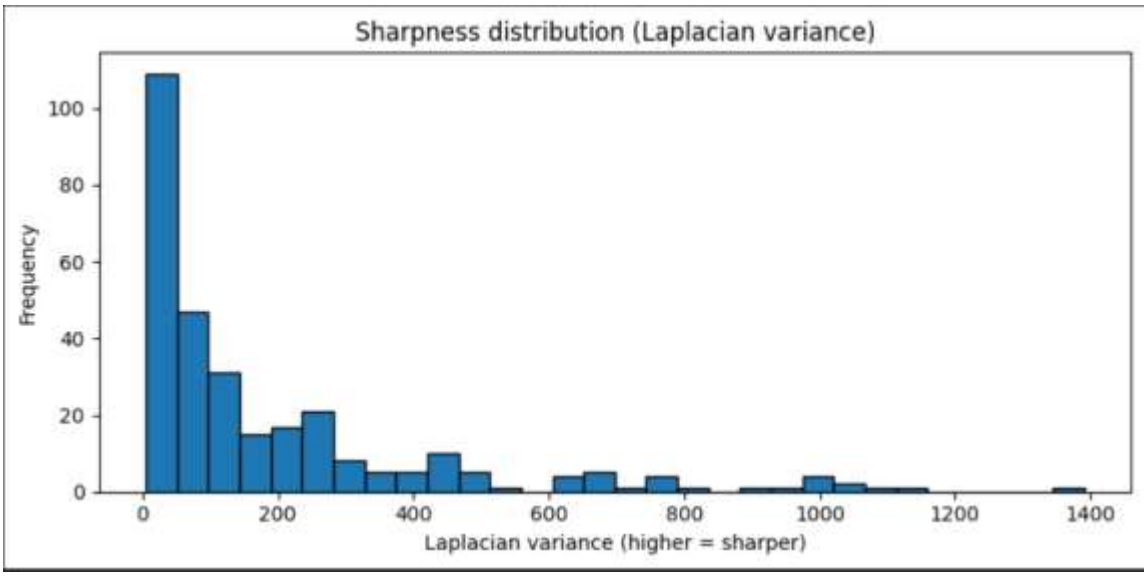

Figure 4.Sharpness Distribution Curve

The sharpness analysis results reveal that most images preserve sufficient clarity of vision for feature extraction of gaze attributes. Consequently, the model training process is successful, and the feature extraction procedure is effective and accurate.

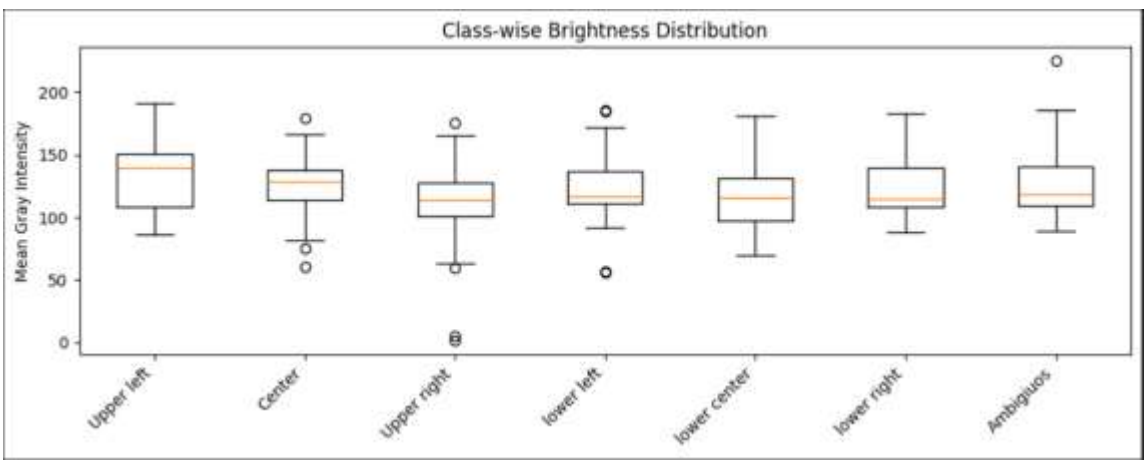

Figure 5.Brightness Distribution by Class

Class-dependent differences in brightness allow realistic image capture scenarios and maintain consistency enough to facilitate proper feature learning. This ensures that the model will generalize well and be robust to changes in lighting and reliably classify gaze directions in different environments.

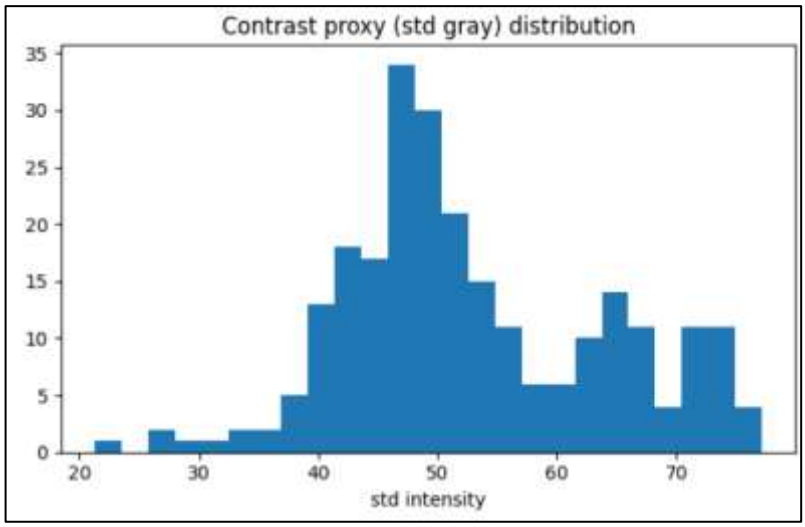

Figure 6. Contrast Proxy Distribution

Analysis through contrast reveals that the visibility of facial features remains intact despite moderate changes in light intensity. This ensures the reliability of feature extraction from images. Consequently, gaze direction prediction becomes more robust.

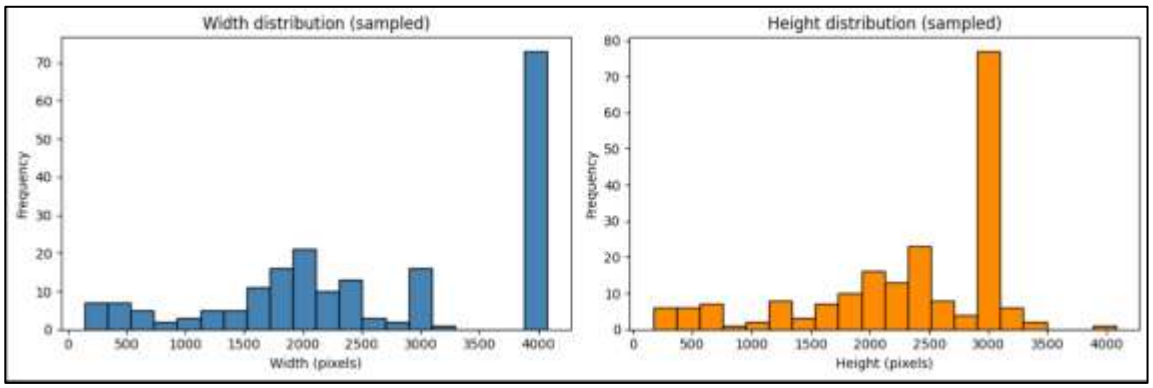

Figure 7. Width and Height Distribution Graphs

The distribution of the images' dimensions verifies consistent preprocessing and standardized inputs within the dataset that are suitable for learning via CNNs. This ensures reliable feature extraction, effective training, and proper classification of gaze directions by minimizing variations caused by differing image dimensions.

The training of the proposed eye gaze classification model is carried out using the ResNet model, and the results are analyzed based on the training and validation performance graphs to evaluate the convergence, learning, and generalization properties of the model.

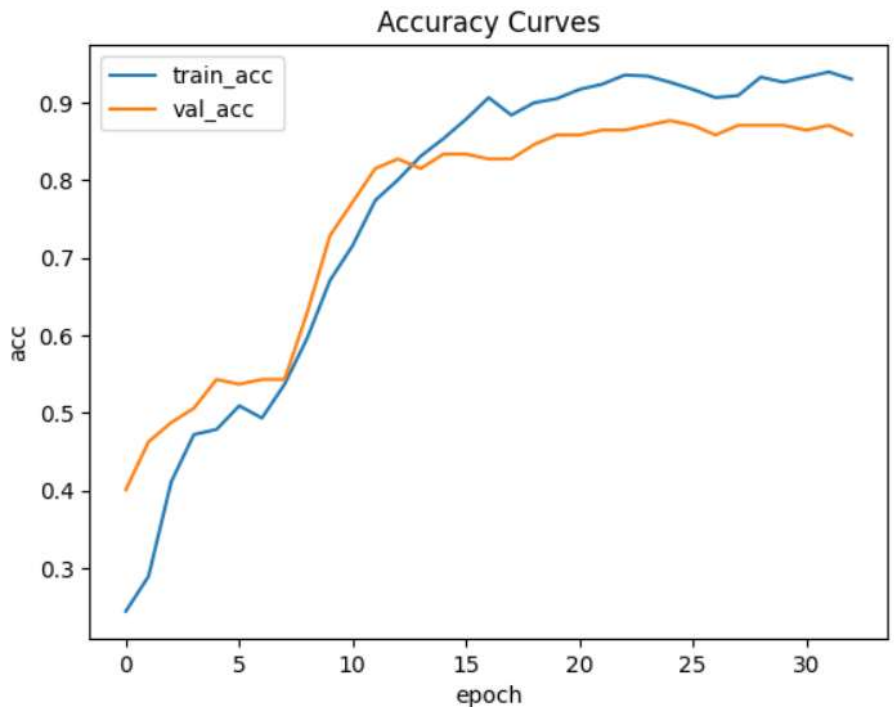

Figure 8. Accuracy Curves

Both training and validation accuracy rose consistently with each epoch, suggesting that the network was able to learn features successfully, optimize effectively, and converge. This shows that the model is capable of generalizing well during the entire process of training.

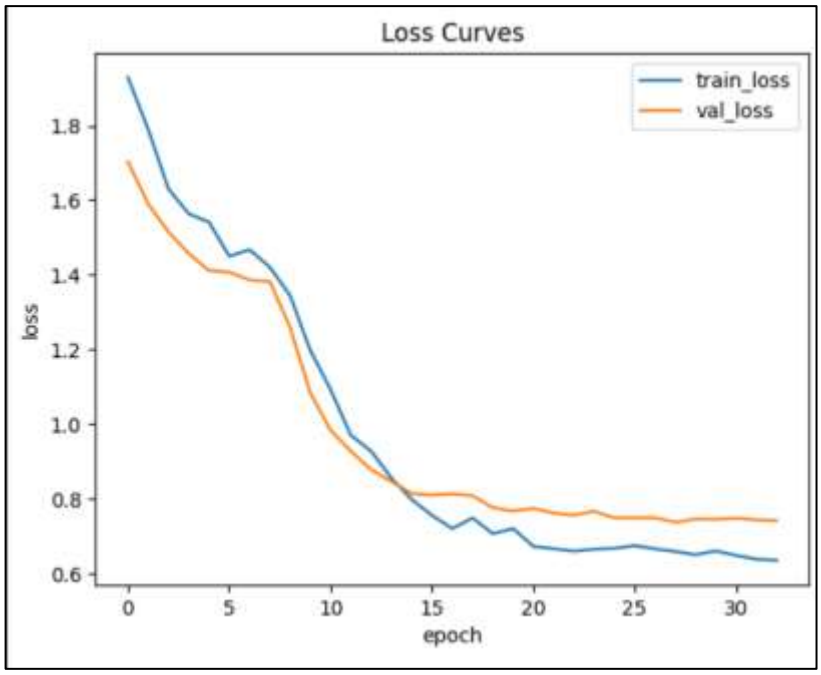

Figure 9.Loss Curves

The falling loss curves suggest that optimization is taking place during training. In addition, since there is close similarity in both training and validation loss curves, it can be inferred that overfitting of the data is not significant in the network.

Performance of the final model on the classification task was determined by employing standard performance measures, such as accuracy, precision, recall, and F1-score, which provided an overall measure of how well the proposed model performed on the classification task. Accuracy gives information about the accuracy of prediction results, while precision and recall measures give more detailed information about class discrimination performance of the model, and F1-score provides a measure of classification performance that combines both precision and recall.

```
Classification report:
              precision    recall  f1-score   support

   Upper left     0.8750    0.8400    0.8571        25
       Center     1.0000    0.9048    0.9500        21
  Upper right     0.9565    0.9565    0.9565        23
   lower left     0.8000    0.8696    0.8333        23
 lower center     0.7778    0.8400    0.8077        25
  lower right     0.8889    0.8889    0.8889        27
    Ambigiuos     0.7778    0.7368    0.7568        19

     accuracy                         0.8650       163
    macro avg     0.8680    0.8624    0.8643       163
 weighted avg     0.8681    0.8650    0.8657       163
```

Figure 10. F1 score

It can be clearly observed from the above classification results that the proposed software-based gaze direction estimation system performed well with respect to all seven gaze areas and had an accuracy rate of 86.50% along with a macro-F1 score of 86.43% and a weighted-F1 score of 86.57%. The results show that the model was able to learn and generalize gaze-related discriminatory characteristics.

The fact that the difference between macro and weighted average performance metrics is minimal signifies balanced behavior across different classes, and thus the model is not biased towards any specific dominant gaze region.

As shown by class-wise analysis, Center and Upper Right gaze directions have demonstrated excellent results, each one reaching an F1-score above 0.95. Such high scores suggest that there is something specific about these gaze directions that makes it easier for the classifier to recognize and classify their distinct visual features. Another gaze direction showing high results is Lower Right.

The relatively low scores of Upper Left, Lower Left, and Lower Center classes imply that such gaze directions have certain similarities or slight overlaps in terms of their directional features, thus increasing the complexity of their recognition and classification. At the same time, despite some difficulties related to these particular classes, their results still remain relatively high enough.

The Ambiguous class obtained the lowest F1-score among all other classes. It is not surprising because of the ambiguous nature of these gaze instances, which means that some of them might contain visual features corresponding to two or more classes

simultaneously. Consequently, there is a higher possibility of ambiguity, which results in lower accuracy in classification.

Overall, from the findings, it is evident that the framework has successfully captured gaze direction patterns through the implementation of a vision-based software algorithm, producing accurate enough predictions that can be used in human-robot cooperation applications. The high accuracy levels of prediction obtained for each direction confirm the viability of employing these predicted gaze areas in controlling the robot's object choice process.

The confusion matrix analysis revealed very good discrimination for most gaze categories, which means that the proposed model would be capable of differentiating gaze regions effectively and making minimal errors in classifying the gaze region. Most correct classifications appear along the main diagonal of the confusion matrix, and it is evident that the accuracy of classifying gaze regions is very high in most cases. Any confusion was witnessed in cases where adjacent and similarly oriented gaze regions were concerned. It makes sense because there are bound to be some overlapping features for such regions.

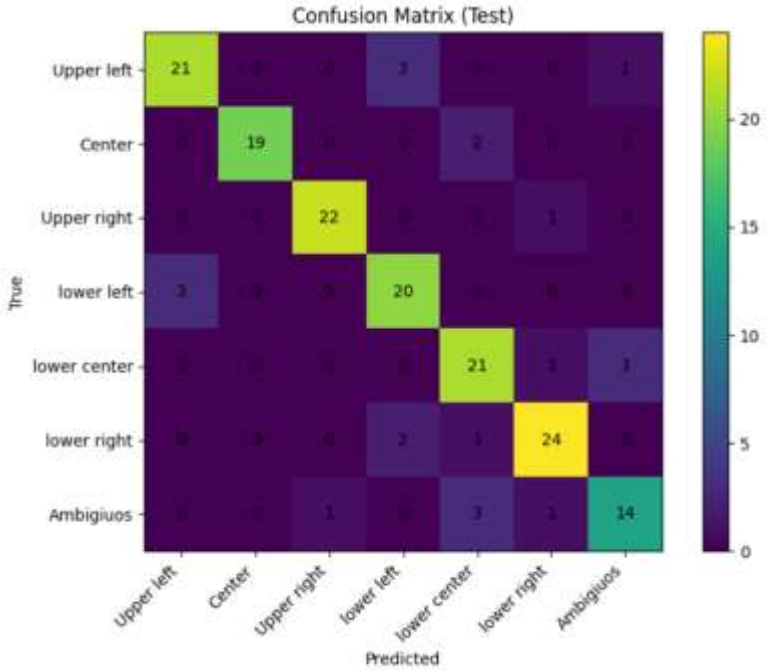

Figure 11.Confusion matrix

Further understanding of the model's behavior regarding gaze classification is provided by the confusion matrix, which shows additional details beyond the average performance

metric statistics. The presence of a pronounced diagonal suggests that the great majority of gaze classes was predicted correctly, reflecting the ability of the algorithm to effectively distinguish all seven gaze regions from each other.

Some gaze regions demonstrate an even stronger correctness of classification with a negligible number of off-diagonal elements, indicating that these classes are distinctive enough to be separated from one another. As could be expected, these are precisely Center, Upper Right, and Lower Right gaze classes with their high precision and F1-scores.

Errors are present at a relatively low level, and mostly take place within neighboring or visually related gaze classes, not distant and completely unrelated to each other. Confusions of Upper Left with Lower Left classes, as well as Lower Center with Lower Right and Ambiguous, can be explained by the fact that neighboring gaze classes have slight overlaps in terms of visual features such as eye position and head orientation.

However, the Ambiguous class appears to show relatively more confusion between neighboring directional classes, especially lower-centered areas, as would be anticipated considering the transitional nature of ambiguous gaze samples. In no way should such high confusion rates be taken to mean model instability but should be regarded as the result of the natural ambiguity found within these gaze samples.

Most importantly, the low levels of confusion in areas far from each other signify that any potential mistakes are localized in nature and are based on actual spatial relations within the data, thus implying that errors are likely to occur in regions close to natural class borders.

As can be clearly seen from the results, the proposed gaze estimation software only approach demonstrates high levels of class separability and robustness under difficult conditions. Such a system proves to be well-suited to be used in gaze-based human-robot collaboration, where accurate prediction of the eye gaze region is required for robotic action.

The recall for each individual class gives an understanding of the effectiveness of the model in detecting samples for each of the gaze classes. These differences show that there are varying levels of sensitivity for each class and can be used to determine the effectiveness of detection for each true gaze sample.

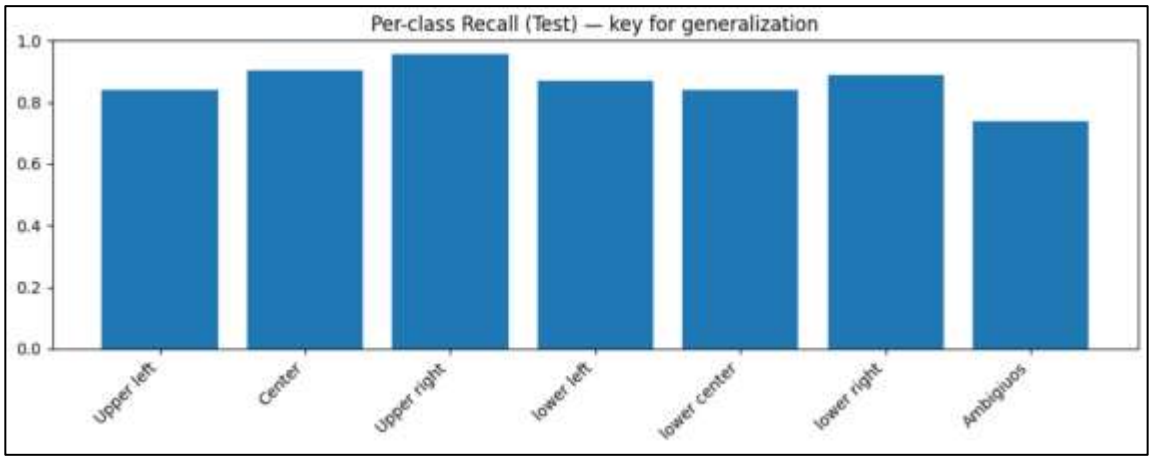

Figure 12. Recall per class(test)

From the class-wise recall values, it can be seen that the proposed model is successful in exhibiting high levels of sensitivity towards most gaze directions, implying that the model performs well in recognizing genuine instances without generating too many false negatives. The higher recall values achieved in gaze directions like Upper Right, Center, and Lower Right imply that the model easily recognizes these directions, while a slight decline in the recall values in gaze directions like Ambiguous and Lower Center is attributed to the complexity of recognizing these directions due to their resemblance to others.

In addition, the proposed model is tested on unseen test data to measure its ability to generalize in a practical environment setting. This will help understand how well the model can predict the gaze direction of unseen samples, thus show casing its feasibility and effectiveness when utilized in human-robot interaction applications.

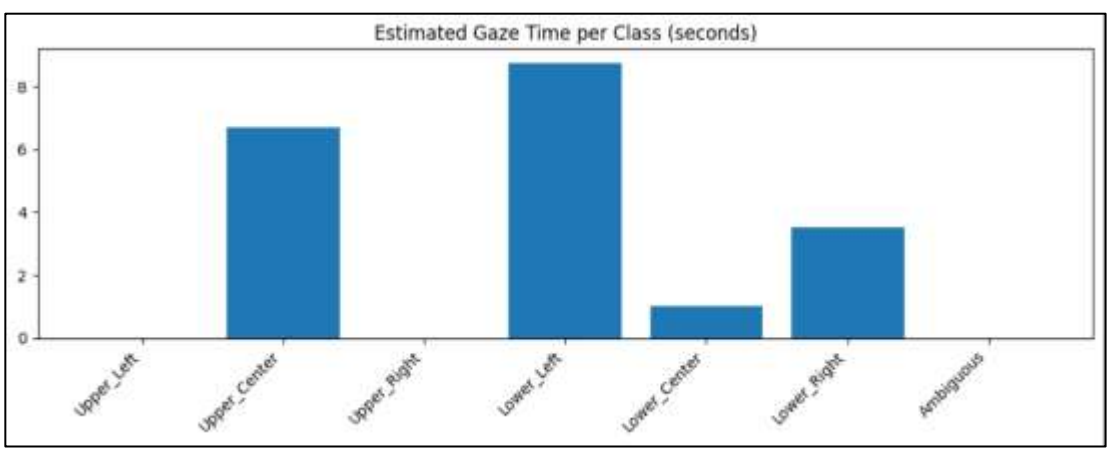

Figure 13. Estimated time per class

This estimate of gaze time per class helps us gain some understanding about the pattern of gaze distribution in different regions and indicates the way in which visual attention could change throughout the interaction process. Long periods of gaze time found in some classes, especially in low and middle regions, could imply a prolonged period of focused attention on these particular target zones, whereas short gaze times in other regions could mean a temporary glance there. This shows that, in addition to being a useful tool for identifying gaze directions, the proposed method is also applicable in analyzing temporal user attention, which would be useful for gaze-based human-robot cooperation.

3.2 Research Findings

This part highlights some important findings obtained through the evaluation of the proposed gaze-guided collaboration between human and robots via software only. From the results of the classification, analysis, and inferences provided in the previous section, the following important findings were realized.

The main conclusion from this study is that it is possible to estimate gaze direction with the help of a software-only solution, using off-the-shelf RGB sensors only. Experimental data shows that accurate gaze classification is possible without using dedicated eye-tracking infrared sensors. Thus, the possibility of using inexpensive vision-based gaze tracking is confirmed by this experiment.

Classification accuracy and class-wise performance show that gaze direction data could be successfully learned by deep neural networks based on regular camera images. This result is important because it proves that the basic idea of this research is viable, and it is possible to build a system for assistive gaze-driven robot-human cooperation without costly hardware. It is also clear that software-only gaze estimation would allow for greater adoption of assistive human-robot collaboration technology because of lower costs.

Another crucial discovery made during the study was the applicability of region-based gaze classification in recognizing intentions and supporting object selection. Instead of predicting continuous gaze coordinates, the developed method allowed successful classification of gaze direction into several predefined gaze regions based on the target object position.

Analysis of the confusion matrix, recall statistics, and class-wise F1 results demonstrates that the classification approach used by the authors facilitated adequate differentiation between various gaze directions while also simplifying robotic control interactions.

This conclusion is significant since it provides evidence to support the applicability of region-based gaze prediction as a viable method for controlling assistive robotics. As an alternative to continuous gaze coordinate calculation, the classification method allows the simplification of robot control while still offering reliable performance in terms of object selection.

A third notable outcome pertains to the efficacy of deep learning-driven feature representation for gaze prediction. According to the results obtained from the experiments, it can be concluded that the ResNet model was able to learn discriminative visual features related to gaze, eye region pattern, and pose, which helped achieve highly accurate classification in most gaze areas. It highlights the potential of the model to identify visual cues necessary for appearance-based gaze prediction, even in varying poses, lighting conditions, and images. The high accuracy achieved in classification suggests that deep visual features learned via the proposed methodology are robust enough to support gaze-based interactions.

The outperforming behavior of the ResNet-18 architecture implies that residual learning is much more effective in modeling gaze-oriented visual cues compared to other network architectures. The hierarchical feature learning abilities of deep residual networks seem to work well with respect to extracting both low-level features as well as higher-level features necessary for predicting gaze direction. The experimental results presented here confirm the appropriateness of using residual deep learning architectures for gaze estimation problems and emphasize the benefits of an automatic feature learning procedure rather than feature engineering. The results also show that learned deep features may contribute to effective perception performance, even without depending on manually crafted features.

Another interesting observation is the varying performance of gaze estimation in different regions, where some of the classes have higher classification performance compared to other classes. High-performing classes like upper-center and upper-right exhibited high discriminability, meaning that these particular gaze directions provide clear visual patterns that the model can detect and classify with high accuracy. The precision, recall, and F1-score values of these classes confirm that the proposed method is capable of accurately estimating the gaze regions with clear direction cues, enabling high classification performance.

On the other hand, classes like ambiguous and lower-center pose relatively higher difficulties compared to other classes, which indicate the higher difficulty level in detecting these particular gaze directions due to the subtle gaze shifts and similarities in visual features between adjacent classes. It can be concluded that all the gaze directions

are not separable at equal degrees; hence, misclassification may occur because of the similar visual patterns that exist in certain regions.

It is important as it not only indicates the practical difficulties involved in estimating gaze but also highlights the possibility for improvement in the area using larger datasets, better pre-processing techniques, or improved modeling. The study also gives us insight into the field of assistive control, wherein gaze regions that pose greater ambiguity would need extra consideration.

In addition to this, another noteworthy result of this research is the fact that the developed model showed encouraging generalization capability when applied to unseen inference samples. Based on the results generated using inference on novel images, it can be inferred that the model is capable of extrapolating beyond the training data and predicting the gaze direction with reliability despite encountering novel inputs. This implies that the representations learned by the model go beyond simple memorization of the training samples and reflect important patterns related to gaze directions that are generalizable to novel instances.

One major key result obtained from the experiment in general is that gaze estimation can be used as an intention channel in collaborative robotic interactions. With the relationship established between gaze prediction and region-based object selection as well as robotic tasks, this work provides a conceptual basis for the practical implementation of using visual intentions for robotic action. This result proves the research hypothesis that gaze may act as a natural interaction mode between humans and robots, especially in assisting actions by different-ability humans. Not just a mere observation tool, gaze was demonstrated to play an important role in guiding robotic actions based on intention. This is among the key contributions of this study.

From the outcomes of the experiments conducted, there arise several important observations from this study. First of all, the success of using only software-based gaze estimation techniques in conjunction with regular RGB cameras proves that such a methodology is suitable for implementing an assistive human–robot collaboration application. Second, it was found out that region-based gaze classification can be used to determine the intention of the user in selecting objects. It must be noted that the high accuracy of ResNet indicates the effectiveness of residual learning in extracting useful gaze-related features from images despite the differences in gaze regions. Moreover, the promising outcomes on unseen inference data indicate the generalizability of the model and ensure its practicability. All of these observations lead to the conclusion that gaze direction can be used as a contactless control method for human–robot collaboration applications.

3.3 Discussion

In this section, we consider the importance of the obtained experimental results and research findings with respect to the research objectives, suggested methodology, and existing literature on the problem of gaze estimation, assistive robotics, and human-robot cooperation. Whereas the previous sections considered the performance results and the research findings obtained, in this section, the emphasis is placed on the interpretation of the meaning and importance of these findings. In particular, in this section, we show that the obtained results provide evidence of the validity of the suggested software-only gaze-guided system for recognizing users' intentions and selecting objects in the context of human-robot cooperation tasks.

Moreover, this section takes into account the possible consequences that arise from the results of the study both for implementation in practice and for future scientific investigations. The results are assessed in conjunction with other related research conducted regarding gaze interaction and assistive human-robot cooperation, pointing out advantages and areas of improvement in the current research. Special emphasis is placed on how the results can influence the implementation of low-cost vision-based gaze control in the area and help in further development of assistive robotics. Thus, apart from reflecting on the significance of the results of this research, the discussion also situates it within the field of gaze-mediated human-robot interaction.

The classification results obtained in this experiment, the developed software-only gaze estimation system has shown high accuracy when performing predictions within the predefined gaze categories, indicating the viability of appearance-based gaze estimation approach based on regular RGB cameras. Classification accuracy, backed up by the performance of class-wise precision, recall and F1-scores, shows that deep learning model learned to distinguish between visually relevant gaze-related characteristics with relatively good robustness.

One of the most interesting results of this experiment is high classification accuracy obtained using ResNet-18 architecture. Such outcome demonstrates the usefulness of residual network architectures in tasks of distinguishing small changes in eye direction and facial appearance that are essential for the performance of gaze estimation tasks.

This conclusion existing works dedicated to deep learning in gaze estimation tasks showing that appearance-based gaze estimation systems based on CNN models can provide high gaze estimation performance even without usage of eye tracking features.

While there were good performances in classification across the whole set, there were inconsistencies in gaze estimation accuracy across different gaze regions. Some regions

such as upper center and upper right gaze regions performed well, while others such as ambiguous and lower center gaze regions were less performant.

This could be explained by the fact that there is similarity among gaze directions in the vicinity and how subtle some gaze shifts can be. Ambiguous gaze samples might have gaze characteristics from both classes, making it more difficult to classify. What this shows is that there is difficulty in separating gaze directions within certain gaze regions. This does not point at a limitation of our proposed framework but rather a problem in gaze estimation in itself. From a research point of view, this is helpful in terms of knowing what can be improved in future work.

Another major finding of this study is that it highlights the possibility of gaze estimation through software alone as an effective substitute for the hardware-dependent gaze tracking systems used in human-machine interactions. The conventional methods of eye tracking involve use of infrared technology along with expensive calibrations and hardware, thereby making it an inaccessible process.On the contrary, the presented system requires only an ordinary RGB imaging camera and advanced machine learning capabilities to estimate the direction of gaze. This opens up the possibility of implementing inexpensive yet highly effective solutions in such applications. The study therefore provides a technically feasible solution with practical significance.

One of the core contributions of this research is in establishing gaze as an effective mode of control as opposed to its role in merely signaling attention. Previous literature often considers gaze as purely observational data that can be employed for monitoring or intent identification purposes. However, the current research incorporates gaze in the process of action selection by the robot. This is critical since it establishes the importance of incorporating gaze beyond merely being a perception cue to command-based control. From the findings, gaze can act as an efficient means of communication between the human operator and the robot, especially in collaborative assistance scenarios where the user may have some disabilities.

The connection between gaze estimation and mapping regions with object choice by a robot has far-reaching effects on collaborative robotics. Whereas existing approaches view human perception and robotic manipulation as separate processes, this system has brought the two into a single intention-to-action process. This is one important aspect of collaborative robotics, whereby the actions of the robot are based on human intention.

Applications could include:

- Assistive robotics
- Collaborative environments
- Smart service robots
- Human-centric automation

These results indicate that gaze-guided interaction might help develop more natural collaborative robotics systems sensitive to user intentions.

Another relevant issue to discuss is that of model generalization. The encouraging results yielded by inference on previously unseen images show that the trained model has successfully learned valuable gaze representation information not simply through memorization of the training data, and therefore has strong generalization properties with respect to the proposed learning paradigm. This is crucial because, to be considered applicable to real-world problems, the assistive human-robot collaboration system requires high-level generalization, meaning that it should be able to cope well with inputs different from those seen during the training phase. On the other hand, since this experiment was based on relatively small and limited data, it is important to take into account that these results can only be regarded as encouraging rather than definitive.

Though there are some positive results in the study, there are several limitations that one needs to keep in mind before drawing conclusions from the data collected during the investigation. Firstly, while the data size used in the study is adequate to develop and test the model, it is small compared to the sizes of large gaze estimation databases available today. Secondly, even though classification results are promising, the performance of the system might still be affected by various conditions such as lighting change, different head poses, and confusion in determining gaze transition points. Lastly, robotic experiments were used merely to show the feasibility of gaze-based selection of an object rather than conducting an extended test of the system in real-life situations. Such limitations do not negate the value of the system created but indicate its scope at this point in time.

From the overall analysis of the results, it is clear that the developed framework successfully establishes the feasibility of using software-only gaze estimation as an intent-based approach for collaborative robot-human interaction. It not only confirms the applicability of the suggested framework but also reveals region-wise performance insights. Although there are some limitations, the conclusions made have implications for the field of gaze estimation as well as collaborative robotics by showing that user visual intent can be exploited by a cost-effective intelligent framework for robotic interaction.

4. Conclusion

In this research, a software-only gaze-directed human-robot collaborative framework was introduced to facilitate the selection of assistive objects based on the user's gaze direction. The main objective of this study was to evaluate the feasibility of employing gaze direction estimation based on traditional RGB cameras and deep learning techniques as an effective control interface for guiding robots in manipulating objects, especially for people with disabilities who need assistive ways of controlling robots.

The motivation for this study was driven by constraints imposed by current gaze-controlled assistive technology, most of which require expensive eye tracking equipment or complicated user interfaces, thereby limiting the scope of their adoption and application. In order to solve this problem, this research proposes a cost-effective vision-driven solution that integrates gaze recognition, intention recognition from regions, and robot manipulation tasks in an assistive human-robot collaborative paradigm.

One of the key contributions of the current work was the development of a software-only gaze direction estimation algorithm with deep learning-based vision perception capability. Thanks to the use of ResNet-18 for the task of region-based gaze classification, it was proved that gaze intention estimation could be performed using camera data only and without additional eye-tracking devices. Classification was performed based on the use of accuracy, F1-score, confusion matrix, and inference tests with satisfactory results.

Another key contribution of the current work was the implementation of a region-based intention map, which linked predictions about gaze directions to pre-specified workspace regions. As a result, gaze estimation was extended beyond the perception problem into the realm of intention, turning visual attention into a command.

The combination of gaze prediction with robotic arm object selection made the contribution of the study even more innovative compared to other gaze tracking studies. The research proves that it is possible to translate the user's visual intention into robot movement within a cooperative environment. Based on the results obtained from the experiments, it can be stated that gaze estimation using only software can provide an efficient way of human–robot interaction based on gaze. The results also indicate that gaze classification based on regions could be a good approach towards simplifying controls.

Furthermore, the study is significant to the wider areas of assistive robotics, gaze estimation, and human-robot collaboration as it illustrates a combined model wherein intelligent sensing and robot movement act as a cohesive intention-to-action mechanism. And also from an assistive technology point of view, the study also reveals the societal

importance of cost-effective robotic gaze-guidance systems to facilitate accessibility, autonomy, and inclusive interactions for users with different abilities.

Overall, the goals of the study have been met successfully through the design, development, and validation of the proposed model. The outcomes confirm the practicality of utilizing gaze as a natural input interface for assistive human-robot collaboration.

Several significant contributions were made through this research. First, it introduced a gaze direction estimation method utilizing solely software tools, based on conventional RGB images and deep neural networks, thus limiting the requirement for expensive eye tracking equipment. Second, it presented a regional intention mapping scheme for integrating gaze estimation and target object selection. Third, an end-to-end gaze-driven robot picking system was created, whereby user vision intention is translated into robotic operation. Fourth, the experiment offers valuable results concerning gaze classification at the region level, model effectiveness, and the viability of gaze-driven human-assistance interaction. In summary, the above contributions offer substantial significance to research on collaborative human-robot assistance.

However, there are some constraints that must be noted regarding this piece of research. Firstly, although the database of images used during this study is enough for developing and testing the proof of concept, it is relatively small for making generalizations for a large scale of cases, which may influence the model's performance. Another constraint that needs to be considered is associated with performance issues related to the influence of environmental factors, such as varying lighting conditions, head positions, and uncertain transitions of gaze. It should be noted that all those problems remain challenging in appearance-based gaze estimation models. Moreover, robotic experiments performed as part of the study only addressed the problem of proving the concept and did not involve users of different abilities.

Some future directions suggested by this work include using gaze datasets that have been expanded by recruiting more participants, increasing the sample size, and incorporating diverse environments during capturing data to improve the generalization capabilities of the model. Another future direction includes implementing more powerful deep learning algorithms than ResNet-18, such as transformers-based vision networks, or combining gaze tracking and other modalities to enhance the precision and feature extraction capabilities of the model. The next promising future direction is advancing the existing region-based technique towards continuous gaze prediction, which will enable improved control of robots and more complex interaction scenarios. Furthermore, multimodal

interaction offers great potential in gaze-based robot control, where gaze can be integrated with speech control, gestures, and adaptive user feedback.

Future developments may also improve the robotic system by incorporating more sophisticated grasping mechanisms, dealing with changing objects in real-time, and having an adaptive collaboration between the user and the robot. In particular, one interesting area of study is to perform user experiments using differently-abled individuals to determine how usable, accessible, and effective the assistive aid actually is in practice. Exploring commercialization and implementation of this type of robotic technology may be useful to help transfer the current research concept to an actual assistive device. In the end, these future research ideas can help bring the present proof of concept to an intelligent and intention-aware collaboration robotic system.

In conclusion, this paper shows that the application of software-based gaze estimation can not only detect the direction of the user's gaze to capture their attention but also direct useful robotic actions as part of a human-robot collaboration system. Thus, combining gaze intentions with robot object manipulation represents a significant contribution towards designing more accessible, intelligent, and human-centric interaction interfaces.It is hoped that this paper will pave the way for future studies and innovations on gaze-directed collaboration between humans and robots.

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